

Problems for week 9

Solutions

Problem 1. Let C denote the triangle with vertices $(0,0)$, $(1,0)$ and $(1,2)$, oriented counterclockwise. Evaluate the line integral

$$\oint_C xy \, dx + x^2 y^3 \, dy$$

by two methods:

- (a) directly;
- (b) using Green's Theorem.

Solution 1.

- (a) Let C_1 be the line segment from $(0,0)$ to $(1,0)$, C_2 be the line segment from $(1,0)$ to $(1,2)$, and C_3 be the line segment from $(1,2)$ to $(0,0)$. Firstly,

$$C_1 : x = t, y = 0 \quad (0 \leq t \leq 1),$$

so $dx = dt$, $dy = 0$, and

$$\oint_{C_1} xy \, dx + x^2 y^3 \, dy = \int_0^1 0 \, dt = 0.$$

Secondly,

$$C_2 : x = 1, y = t \quad (0 \leq t \leq 2),$$

so $dx = 0$, $dy = dt$, and

$$\oint_{C_2} xy \, dx + x^2 y^3 \, dy = \int_0^2 t^3 \, dt = \left[\frac{t^4}{4} \right]_0^2 = 4.$$

Thirdly, $-C_3$ i.e. C_3 backwards from $(0,0)$ to $(1,2)$ is given by

$$-C_3 : x = t, y = 2t \quad (0 \leq t \leq 1),$$

so $dx = dt$, $dy = 2dt$, and

$$\oint_{-C_3} xy \, dx + x^2 y^3 \, dy = \int_0^1 (2t^2 + 16t^5) \, dt = \left[\frac{2t^3}{3} + \frac{8t^6}{3} \right]_0^1 = \frac{10}{3}.$$

So¹ $\oint_{C_3} = -\frac{10}{3}$. Finally,

$$\oint_C xy \, dx + x^2 y^3 \, dy = \oint_{C_1} + \oint_{C_2} + \oint_{C_3} = 0 + 4 - \frac{10}{3} = \frac{2}{3}.$$

¹Without switching the orientation of C_3 , one could parametrise it as $x = 1 - t$, $y = 2 - 2t$, $0 \leq t \leq 1$; this yields the same answer.

(b) By Green's Theorem,

$$\oint_C xy \, dx + x^2 y^3 \, dy = \iint_R \left(\frac{\partial}{\partial x}(x^2 y^3) - \frac{\partial}{\partial y}(xy) \right) \, dA,$$

where R is the region enclosed by C . Thus

$$\oint_C xy \, dx + x^2 y^3 \, dy = \iint_R (2xy^3 - x) \, dA = \int_0^1 \int_0^{2x} (2xy^3 - x) \, dy \, dx.$$

Inner integral:

$$\int_0^{2x} (2xy^3 - x) \, dy = \left[\frac{xy^4}{2} - xy \right]_0^{2x} = 8x^5 - 2x^2.$$

Outer integral:

$$\int_0^1 (8x^5 - 2x^2) \, dx = \left[\frac{4x^6}{3} - \frac{2x^3}{3} \right]_0^1 = \frac{2}{3}.$$

Problem 2. Let $\mathbf{F} = (y - \cos y, x \sin y)$ and let C be the circle $(x - 3)^2 + (y + 4)^2 = 4$ oriented clockwise. Use Green's Theorem to evaluate the line integral

$$\oint_C \mathbf{F} \cdot d\mathbf{r}.$$

Solution 2. C is clockwise and encloses the disc R of radius 2 centred at $(3, -4)$. Therefore

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= - \oint_{-C} \mathbf{F} \cdot d\mathbf{r} = - \iint_R \left(\frac{\partial}{\partial x}(x \sin y) - \frac{\partial}{\partial y}(y - \cos y) \right) \, dA \\ &= - \iint_R -1 \, dA = \iint_R 1 \, dA = \text{area}(R) = \pi 2^2 = 4\pi. \end{aligned}$$

Problem 3. If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, then $f\mathbf{F}$ is the vector field defined by

$$(f\mathbf{F})(x, y) := f(x, y)\mathbf{F}(x, y).$$

Assume that the appropriate partial derivatives exist and are continuous. Show that

$$\text{div}(f\mathbf{F}) = f \text{div}(\mathbf{F}) + \mathbf{F} \cdot \nabla f.$$

Solution 3. If $\mathbf{F} = (P, Q)$, then

$$\begin{aligned} \text{div}(f\mathbf{F}) &= \frac{\partial(fP)}{\partial x} + \frac{\partial(fQ)}{\partial y} = (fP_x + f_x P) + (fQ_y + f_y Q) \\ &= f(P_x + Q_y) + (P, Q) \cdot (f_x, f_y) = f \text{div}(\mathbf{F}) + \mathbf{F} \cdot \nabla f. \end{aligned}$$

Problem 4. Compute the area of the region enclosed by the *cardioid* curve, parametrised by

$$C : (x(t), y(t)) = (-\cos(2t) + 2\cos t, -\sin(2t) + 2\sin t), \quad 0 \leq t \leq 2\pi.$$

Solution 4. Let $R \subset \mathbb{R}^2$ be the region in question:

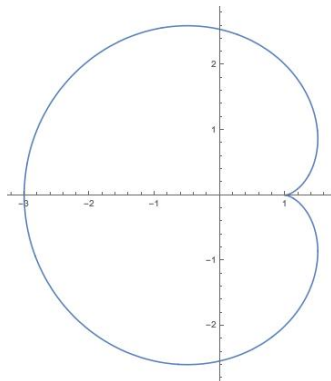


FIGURE 1. The cardioid parametrised by $x(t) = -\cos(2t) + 2\cos t$ and $y(t) = -\sin(2t) + 2\sin t$, for $0 \leq t \leq 2\pi$.

Since

$$(x'(t), y'(t)) = 2(\sin(2t) - \sin t, -\cos(2t) + \cos t),$$

we have by Green's Theorem that

$$\begin{aligned} \text{area}(R) &= \frac{1}{2} \oint_C x \, dy - y \, dx = \frac{1}{2} \int_0^{2\pi} x(t)y'(t) \, dt - y(t)x'(t) \, dt \\ &= \int_0^{2\pi} (-\cos(2t) + 2\cos t)(-\cos(2t) + \cos t) \, dt - (-\sin(2t) + 2\sin t)(\sin(2t) - \sin t) \, dt \\ &= \int_0^{2\pi} (3 - 3(\cos t \cos(2t) + \sin t \sin(2t))) \, dt = \int_0^{2\pi} 3 \, dt = 6\pi. \end{aligned}$$

In the penultimate equality we used the facts that

$$\int_0^{2\pi} \cos t \cos(2t) \, dt = \int_0^{2\pi} \sin t \sin(2t) \, dt = 0,$$

which can be verified e.g. by integration by parts or by invoking appropriate product-to-sum formulae.

A function $g : R \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is called *harmonic* on R if it satisfies Laplace's equation $\Delta g = 0$ on R , where

$$\Delta g := \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2}.$$

Problem 5. Use Green's Theorem in Normal Form to prove **Green's first identity**:

$$\iint_R f \Delta g \, dA = \oint_C f(\nabla g) \cdot \hat{n} \, ds - \iint_R \nabla f \cdot \nabla g \, dA,$$

where R and C satisfy the hypotheses of Green's Theorem and the appropriate partial derivatives of f and g exist and are continuous. (The quantity $\nabla g \cdot \hat{n} = D_{\hat{n}}g$ occurs in the line integral. This is the directional derivative in the direction of the normal vector $\hat{n}$ and is called the *normal derivative* of g .)

Solution 5. Using Green's Theorem in Normal Form, and the result in Problem 3 (i.e., if $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ have continuous partial derivatives, then $\operatorname{div}(f\mathbf{F}) = f\operatorname{div}(\mathbf{F}) + \mathbf{F} \cdot \nabla f$), we have that

$$\oint_C f(\nabla g) \cdot \hat{n} \, ds = \iint_R \operatorname{div}(f\nabla g) \, dA = \iint_R (f\operatorname{div}(\nabla g) + \nabla g \cdot \nabla f) \, dA.$$

To finish, note that

$$\operatorname{div}(\nabla g) = \operatorname{div}\left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{\partial g}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial g}{\partial y}\right) = \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} = \Delta g.$$

Problem 6. Use Green's first identity to prove **Green's second identity**:

$$\iint_R (f\Delta g - g\Delta f) \, dA = \oint_C (f\nabla g - g\nabla f) \cdot \hat{n} \, ds$$

where R and C satisfy the hypotheses of Green's Theorem and the appropriate partial derivatives of f and g exist and are continuous.

Solution 6. From Green's first identity, we have that

$$(1) \quad \iint_R f \Delta g \, dA = \oint_C f(\nabla g) \cdot \hat{n} \, ds - \iint_R \nabla f \cdot \nabla g \, dA,$$

and that

$$(2) \quad \iint_R g \Delta f \, dA = \oint_C g(\nabla f) \cdot \hat{n} \, ds - \iint_R \nabla g \cdot \nabla f \, dA$$

(just by reversing the roles of f and g). The desired result follows from subtracting (2) from (1).